Examining Equilibrium Nuclear Burning

Consider the CN cycle reactions[1]:

1) $C^{12}+H^1 \Rightarrow C^{13}$ $S_{12,1} = 1.34$ keV-b
2) $C^{13}+H^1 \Rightarrow N^{14}$ $S_{13,1} = 7.60$ keV-b
3) $N^{14}+H^1 \Rightarrow N^{15}$ $S_{14,1} = 1.66$ keV-b
4) $N^{15}+H^1 \Rightarrow C^{12}+He^4$ $S_{15,1} = 7.3\text{x}10^4$ keV-b

Express the changes in the mass fractions as:

$$\frac{\partial X_l}{\partial t} = \pm A_l R_{jk} \frac{X_j X_k}{1+\delta_{jk}}$$

(Eqns 5.44-5.47 in the textbook[2]) where *l* represents the species of interest, and *j* and *k* the reactants in reactions that destroy or produce element *l* (note that occasionally, *j*=*k*) the sums represent reactions that create and destroy the element of interest. Use the definitions of R_{jk} (Eqn. 5.47) and K_{jk} (Eqn. 5.33) in the book, with $\tau_{jk} = 19.72\left(\frac{W_{jk}}{T_7}\right)^{1/3}$, $W_{jk} = Z_j^2 Z_k^2 \frac{A_j A_k}{A_j + A_k}$.

1.) Let's examine the CN cycle analytically:
   a. Write down the equations for the changes in mass fraction of elements modified by the CN cycle (H, He^4, C^{12}, C^{13}, N^{14}, N^{15}) for the four reactions above; you can neglect the $N^{15}+H^1 \Rightarrow O^{16}$ branch (ON branch) for the purposes of this question. *Which of these equations do you expect to come into equilibrium, given enough time on the main sequence?*
   b. You can adopt reaction rates of the form above where A = atomic mass, Z = atomic number, and T_7 = temperature in units of 10^7 K. Compute the W and τ values for the 4 different CN cycle reactions above for a *temperature of the solar core at 4.57 Gyr* from the "Test_solarGS98_base model" in the YREC test suite.
   c. Derive expressions for the equilibrium abundances for the CN cycle alone at this temperature (neglecting the ON channel). You should calculate abundance ratios (X_i/X_j), and then use number density normalization condition $X_{CNtot} = \sum X_{12C} + X_{13C} + X_{14N} + X_{15N}$ to express abundances as a function of X_{CNtot}. In addition, use these equations to explain:
      i. why the C^{12} to C^{13} ratio in equilibrium is only weakly temperature dependent; and
      ii. why the net effect of the CN cycle is to transform most of the carbon into nitrogen.

2.) Now, let's take a look at the numerical behavior in YREC. You can use the "Test_solarGS98_base" test suite case. In this problem, it will be useful to look both at the summary of stellar properties as a function of time (.track file) and profiles of properties as a function of radius/mass at a given time (.store file). You may want to

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[1] We're deliberately condensing the $N^{13} \rightarrow C^{13}+e^+ + \nu_e$ and $O^{15} \rightarrow N^{15}+e^+ + \nu_e$ decays into their preceding reactions, because both happen extremely quickly.

[2] Marc H. Pinsonneault and Barbara S. Ryden, Stellar Structure and Evolution (Cambridge: Cambridge University Press, 2023)

increase the number of output profiles over the example in the test suite using NPRTMOD in nml1.

d. Look at the terminal profile at solar age from the .store file. How is energy being transported in different regions of the star? Where is the nuclear burning occurring, and over what fraction of the star?

e. Look at values in the .track file over the solar lifetime. How does the luminosity from CNO burning compare to that from pp chain burning today? How has it changed with time? If you focus on the last 2 Gyrs of the solar history, what is driving the slow change in the relative importance of CNO burning with time?

f. What is the $^{14}N/(^{12}C+^{13}C+^{14}N+^{15}N)$ ratio in the core in the 4.57 Gyr solar model? How does it compare to our analytic value?

g. There is a sharp feature in the CNO burning relative to pp chain burning at early times. Given what you know about the reactions in the CN chain, what causes this transient feature? (If you're stuck, I suggest making plots of the mass fractions of the various CNO components that YREC is tracking as a function of time, and perhaps looking at some profiles at interesting moments in the evolution with respect to this feature.). What assumption in Part 1 is violated at these early times? How does this enhanced CNO burning affect the structure of the star?

h. Finally, let's look at masses other than solar. You can select "Test_mXpXX_feh0p0_base_TAMS" cases from the test suite as a starting place. Examine models in the range of 1.0 to 2.0 $M_\odot$

    i. Construct a plot that shows the ratio of the luminosity generated by CNO and the pp burning chains as a function of mass. For each mass, plot this ratio at the moment when the star has reached a central hydrogen mass fraction of $X_{core,H1} = 0.4$.
    ii. Remake CNO/pp luminosity ratio plot, but now use the central temperature as the x coordinate, instead of mass (i.e. the central temperature for mass M when its core hydrogen fraction = 0.4). At what core temperature does CNO become the dominant source of luminosity over pp burning?
    iii. Use the same models at $X_{core,H1} = 0.4$, but now plot the fractional mass of the convective core (M_{cz}/M_{star}) as a function of mass and central temperature. How do these plots compare to plots examining the importance of the burning chains? Physically, what produces the correlation between CNO burning and convective core mass?